gotten far, Einstein's memoir came into my hands and set me afire" (quoted in [**52**, p. 62]). The inflaming memoir was Einstein's account of general relativity.

In 1917 Weyl gave a lecture course on general relativity at the ETH and soon turned it into the widely read book *Space-Time-Matter*, which went through four different editions within five years and was admired by Einstein himself as a "symphonic masterpiece" (quoted in [**46**, p. 65]). Weyl, however, could hardly resist the impulse to improve on Einstein's belabored mathematics. He was convinced that Riemannian geometry, on which Einstein based his theory, was not a consistently infinitesimal geometry and set out to create a "purely infinitesimal geometry". When he explained to a student that in Riemannian geometry the direction of the transported vector depended on the path, the student innocently asked whether the vector length changed as well. "Of course I gave him the orthodox answer at that moment, but in my bosom gnawed the doubt", Weyl recalled (quoted in [**52**, p. 154]). The orthodox answer was *no*, but he was tempted to see what would happen if the length indeed depended on the path. Weyl questioned the assumption of a fixed distance scale, or "gauge", implicit in Riemannian geometry, and arrived at a more general geometry, in which the gauge factor varied from point to point in space-time, just as railway gauge varied from country to country in the early 20th century [**42**, p. 3].

Weyl aspired at a unified field theory, bringing together the gravitational and the electromagnetic fields. Einstein was delighted to count Weyl among the supporters of general relativity but was less than enthusiastic to see him compete in the construction of physical theories. Einstein praised Weyl's theory as a "stroke of genius of the highest order" and concluded that "except for the agreement with reality it is in any case a grandiose achievement of thought" (quoted in [**52**, pp. 163–164]). As a true mathematician, Weyl could hardly share Einstein's obsession with "reality". As he later argued, "[I]t becomes evident that now the words 'in reality' must be put between quotation marks; we have a symbolic construction, but nothing which we could seriously pretend to be the true real world" (quoted in [**42**, p. 15]). In his reply to Einstein, Weyl pointedly wrote: "It must be